

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders

Game Based Dysgraphia intervention module

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Project Proposal Report

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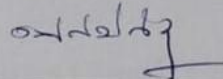
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Dysgraphia is a common neurodevelopmental learning disorder that affects children's ability to produce clear and organized written expression despite having normal intelligence and adequate classroom instruction. Children with dysgraphia often experience persistent difficulties in letter formation, spacing, alignment, spelling, and writing fluency, which can negatively affect their academic performance, classroom participation, and self-confidence. In Sri Lanka, early identification and intervention for Sinhala-medium children with handwriting difficulties are often delayed due to limited access to trained specialists, high clinical assessment costs, and the absence of language-specific digital intervention tools. Additionally, therapists and parents frequently face challenges in guiding handwriting practice at home, reducing opportunities for consistent intervention outside clinical or school environments.

To address these limitations, this research proposes a game-based adaptive web-based mobile handwriting intervention system designed for Sinhala-speaking children aged 6–8 years. The system initially identifies the child's handwriting difficulty level through progressive handwriting game activities and analyzes handwriting characteristics such as letter formation accuracy, spacing consistency, alignment deviation, and letter mirroring. Based on these metrics classifies the child's difficulty level into mild, moderate, or severe categories and recommends personalized intervention activities. The platform integrates interactive handwriting games, sensory-based learning tasks such as writing letters in sand, and voice-guided instructions in Sinhala to support young learners who may have limited reading ability. The system also provides progress monitoring dashboards that enable parents, teachers, and therapists to track improvement trends and identify areas requiring further support.

The application is developed using React and React-Konva for the interactive frontend, Node.js with Express for backend services, MongoDB and Firebase for data management. The system incorporates a hybrid handwriting verification approach by combining OpenCV with a machine learning (ML) model for sensory handwriting image analysis at both letter and word levels, along with MyScript handwriting recognition for word correctness validation and Text-to-Speech (TTS) technology for voice-guided instructions. Expected outcomes include measurable improvements in handwriting accuracy, spacing consistency, and writing fluency, along with increased learner engagement and stronger parental and therapist involvement. Socially, the system promotes equitable access to early handwriting intervention for Sinhala-speaking children, while commercially it offers a scalable and cost-effective digital tool for educational and therapeutic institutions.

Keywords: Dysgraphia, Adaptive Learning, Game-Based Intervention, Handwriting Analysis, Mobile Health Technology

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LIST OF ABBREVIATIONS

Abbreviation	Word/Description
NLDs	Neurodevelopmental learning disorders
SDGs	Sustainable Development Goals
AR	Augmented Reality
DSR	Design Science Research
UI	User Interface
VS Code	Visual Studio Code
IDE	integrated development environment
IP	Intellectual Property
TTS	Text-to-Speech

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1.INTRODUCTION

1.1 Background and Context

Dysgraphia is one of the common neurodevelopmental learning disorders that affects written expression and handwriting abilities in school-aged children. It is characterized by persistent difficulties in letter formation, spacing, alignment, spelling, and writing fluency, even among children with normal intelligence and adequate classroom instruction. Children with dysgraphia often struggle to produce legible handwriting, maintain consistent spacing between letters and words, organize written thoughts, and write at an appropriate speed. Since written communication plays a crucial role in academic learning and classroom assessments, handwriting difficulties can significantly affect a child's educational progress, academic performance, and self-confidence [1].

Research indicates that children aged 6–10 years represent a critical stage for developing foundational handwriting and written language skills [2]. If appropriate support is delayed during this stage, children with dysgraphia may experience persistent handwriting difficulties, slower academic progress, frustration with writing tasks, and reduced motivation to participate in classroom activities. These challenges can also negatively affect children's emotional well-being and classroom participation. Early identification and intervention are therefore essential to support children's academic development and prevent long-term learning difficulties.

Globally, studies estimate that 5–20% of school-aged children experience learning disorders, including dysgraphia and related writing difficulties [1]. In many low-resource and underserved regions, including rural areas of Sri Lanka, children with handwriting difficulties often remain undiagnosed or receive delayed intervention due to limited awareness, high costs associated with clinical assessments, and a shortage of trained specialists such as occupational therapists and educational psychologists [3]. Furthermore, most primary school students in Sri Lanka are educated in the Sinhala language, which contains complex character structures and curved strokes that may pose additional challenges for children experiencing handwriting difficulties.

Existing intervention approaches include therapist-guided occupational therapy, handwriting practice exercises, and specialized learning programs designed to improve fine motor skills and writing accuracy. While some digital handwriting applications are available, most focus primarily on simple letter tracing activities without offering comprehensive systems for structured assessment, adaptive learning, or long-term progress monitoring tailored specifically to dysgraphia-related handwriting challenges.

In addition, many existing digital tools are designed primarily for English, which limits their effectiveness for Sinhala-medium learners. Therapists also frequently face challenges in guiding children's handwriting practice at home, which reduces the effectiveness of intervention outside clinical or therapy sessions. As a result, there is a need for accessible, engaging, and language-appropriate digital solutions that can support both clinical guidance and home-based practice.

To address these limitations, this research proposes a game-based adaptive handwriting intervention system for Sinhala-medium children aged 6–8 years, designed to support both therapist-guided and home-based learning environments.

The proposed system includes the following components:

Initial Level Identification:

Children begin with handwriting games designed for lower levels and gradually progress through the activities. The system identifies the stage at which they experience difficulty completing tasks. It also analyzes handwriting characteristics such as spacing consistency, alignment errors, and letter mirroring to classify the child's difficulty level as mild, moderate, or severe.

Sensory-Based Writing Activities:

The system recommends surface-touching learning activities, such as writing letters in sand, to support sensory-based learning. The child's handwritten work is captured as an image and analyzed to determine whether the letters are written correctly. These activities are incorporated because sensory learning helps children recognize and remember letter shapes more effectively through touch and movement, strengthening motor-memory connections.

Progress Monitoring and Reassessment:

After predefined intervention periods, the system reassesses handwriting performance and provides visual dashboards for parents, teachers, and therapists to monitor improvement trends and identify areas that require further support.

This solution aligns with the United Nations Sustainable Development Goals (SDGs). It supports SDG 4: Quality Education by promoting inclusive and equitable learning opportunities for children with learning difficulties. It also contributes to SDG 3: Good Health and Well-Being by supporting cognitive and emotional development through early intervention. Additionally, it addresses SDG 10: Reduced Inequalities by providing accessible digital support for children in underserved and resource-limited communities [4].

The proposed system falls under the FoC Research Cluster: Technology Integration & Management, integrating Educational Technology (EdTech) and digital health solutions to develop a scalable, intelligent, and accessible intervention platform for children with neurodevelopmental learning disorders (NLDs).

1.2 Literature Review

Digital technologies are increasingly explored as intervention tools for children with dysgraphia. Dysgraphia is characterized by persistent difficulties in handwriting, including poor letter formation, inconsistent spacing, alignment errors, and reduced writing fluency.

Kariyawasam et al. proposed a gamified screening and intervention system for dyslexia, dysgraphia, and dyscalculia, focusing on improving engagement in early learning activities through interactive gameplay elements [5]. Their system integrates educational games designed to detect and support children with learning disabilities using task-based activities and reward mechanisms. Experimental results indicated that gamified learning environments increased student motivation and participation. However, the proposed system primarily focused on screening and basic intervention, without incorporating detailed handwriting performance analysis or adaptive learning pathways tailored specifically for dysgraphia-related handwriting difficulties.

Avishka et al. developed a mobile application designed to support individuals with dyslexia and dysgraphia using assistive learning techniques and interactive exercises [6]. The application included reading assistance, writing guidance, and phonetic support to help learners overcome literacy challenges. The study demonstrated that mobile technologies can provide accessible support for individuals with learning disabilities. However, the system focused mainly on general learning support, and it did not include detailed handwriting analytics, structured therapy progression, or continuous progress monitoring features.

Berninger et al. examined handwriting difficulties through structured writing tasks and emphasized the importance of distinguishing motor execution deficits from higher-level orthographic processing issues [7]. Their findings showed that children with dysgraphia demonstrate measurable deficits in letter formation speed and spatial organization. However, the assessment procedures relied heavily on manual scoring and specialist observation.

Tablet-assisted handwriting applications such as Sinhala- Follow and Writing Wizard have been widely used to improve letter formation and fine motor coordination [8]. These applications typically employ touch-based tracing exercises with visual and auditory feedback. Experimental evaluations report improvements in motor control, letter recognition, and writing confidence among early learners. However, these tools primarily emphasize repetitive tracing activities.

Raza et al. developed an interactive mobile application for assessing phonological and surface dysgraphia in children aged 5-12 using MyScript handwriting recognition technology [9]. The system incorporated a structured 20-word writing test and provided real-time feedback. Preliminary testing demonstrated usability and engagement among participants. Nevertheless, the application focused primarily on diagnostic assessment of two dysgraphia subtypes, structured therapeutic progression, or longitudinal progress tracking.

Rosenblum investigated digitizing tablets for handwriting analysis using kinematic features such as pressure, velocity, and stroke duration [10]. The study demonstrated that digital handwriting analysis can objectively identify motor irregularities in children with dysgraphia.

While this approach provides precise measurement, it requires specialized hardware and does not integrate adaptive therapeutic interventions.

Wan et al. applied deep learning models to analyze handwriting dynamics for dysgraphia detection using digitizing pen features such as velocity, stroke pressure, and temporal patterns [11]. Their results demonstrated high classification accuracy, indicating the promise of machine learning for objective identification. However, the system focused solely on detection and did not provide adaptive intervention pathways for improving handwriting skills.

Coupe et al. developed tablet-based adaptive handwriting games aimed at personalized dysgraphia remediation [12]. Their approach combined engagement mechanics with performance tracking but remained an early prototype without long-term validation or integrated parent profiles and dashboards.

Game-based educational platforms have also been studied for improving writing engagement. Studies on gamification in literacy interventions show increased motivation and task persistence when reward-based mechanics are incorporated [13]. However, most gamified writing systems lack deep personalization mechanisms.

Overall, existing literature confirms that digital handwriting tools can support engagement and motor improvement. However, most systems focus either on assessment, tracing practice, or classification, rather than integrated adaptive therapeutic intervention with continuous monitoring and stakeholder dashboards.

1.3 Research Gap

1. Lack of Language-Specific Dysgraphia Intervention Tools for Sinhala [9]

Most existing digital handwriting intervention systems and handwriting recognition applications are mainly designed for the English language. Although many mobile handwriting learning apps help improve literacy skills, they rarely support languages with unique writing systems. Sinhala writing has special characteristics such as complex curves, combined letters, and specific stroke patterns. As a result, existing handwriting applications and recognition frameworks are not well optimized for Sinhala handwriting patterns. This creates a significant gap in the availability of specialized digital intervention tools designed specifically for Sinhala-speaking children with dysgraphia.

2. Emphasis on Detection Rather Than Structured Intervention [5],[11], [12]

Recent research has demonstrated the effectiveness of artificial intelligence and handwriting dynamic analysis in detecting dysgraphia. Studies using deep learning models and digitizing pen data have achieved high accuracy in identifying handwriting irregularities [11], [12]. However, these systems primarily focus on detection and classification rather than implementing structured therapeutic interventions that help children progressively improve their handwriting skills. As a result, a significant gap exists between automatic dysgraphia identification and practical intervention systems that guide children through structured handwriting improvement activities.

3. Lack of Integration of Sensory-Based Learning Approaches in Digital Intervention Systems

Research in occupational therapy highlights the effectiveness of sensory-based learning techniques, such as tactile writing activities, in helping children with handwriting difficulties improve letter recognition and motor coordination. However, most digital handwriting intervention systems focus primarily on screen-based tracing exercises without integrating sensory-based learning approaches. As a result, children may not receive the full benefits of multi-sensory learning experiences that combine touch, movement, and visual feedback to strengthen motor-memory connections involved in handwriting.

4. Limited Adaptive and Therapeutic Intervention Mechanisms [8], [9], [13]

Many existing handwriting learning applications, including tablet-based tracing tools such as Sinhala-Follow and Writing Wizard, mainly rely on fixed tracing exercises and predefined difficulty levels [8]. While these tools help children practice letter formation and improve motor coordination, they do not dynamically adapt learning tasks based on the child's specific handwriting difficulties. In addition, these applications do not focus on improving the child's particular struggling areas by providing similar, simple practice tasks to strengthen those skills. Most existing tools also emphasize repetitive skill practice, engagement, or screening rather than structured therapeutic intervention.

5. Lack of Voice-Guided Instruction [8]

Most existing handwriting learning applications rely primarily on text-based instructions to guide children through writing tasks and game-based exercises. However, children in the early primary age group (6–8 years) often have limited reading ability and may struggle to understand written instructions independently. As a result, they may experience difficulty understanding how to complete specific handwriting tasks or interact with game-based activities. Current systems rarely provide voice-guided instructions that clearly explain the task steps in a child-friendly manner. The absence of audio guidance can reduce accessibility and engagement for young learners, particularly those with dysgraphia who may already experience frustration during writing activities.

6. Lack of Continuous Progress Monitoring and Dashboards [10], [8]

Although digitizing tablets can capture handwriting kinematics [10], most systems do not convert this data into dashboards for parents and teachers. Existing tracing apps focus on task completion rather than longitudinal tracking [8], limiting the ability of teachers and parents to monitor progress, detect improvement trends, or identify persistent writing difficulties.

Justify novelty and creativity

To address the limitations identified in existing studies, this research proposes a novel Sinhala language-specific dysgraphia intervention system that integrates sensory-based handwriting activities, hybrid handwriting verification, and voice-guided learning within a single adaptive platform. A key innovation is the inclusion of sensory-based writing tasks, such as writing letters in sand, where handwritten input is captured and analyzed to verify correctness, enhancing motor-memory development through tactile interaction.

The system also introduces a hybrid handwriting verification mechanism by combining OpenCV-based image preprocessing, a machine learning model for letter-level analysis, and MyScript recognition for word-level validation. This approach improves accuracy and robustness when handling variations in children’s handwriting.

In addition, Sinhala voice-guided instructions improve accessibility and engagement for young learners. The system further analyzes handwriting features such as letter formation, spacing, alignment, and mirroring to identify the child’s difficulty level and provide adaptive, personalized intervention. A progress monitoring dashboard enables parents, teachers, and therapists to track improvement over time.

Overall, the proposed system demonstrates novelty by integrating sensory learning, hybrid verification, Sinhala voice interaction, and adaptive game-based intervention into a unified platform for supporting Sinhala-speaking children with dysgraphia.

Feature	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	Proposed System
Language-specific support (Sinhala)	✓ Limited	✓ Limited	✗	✓	✗	✗	✗	✗	✓
Voice-guided instructions	✗	✗	✗	✗	✗	✗	✗	✗	✓
Sensory-based learning activities	✗	✗	✓ Limited	✗	✗	✗	✗	✗	✓
Adaptive difficulty mechanism	✗	✗	✗	✓ Limited	✗	✗	✗	✓ Limited	✓
Multi-dimensional handwriting analysis (spacing, alignment, pressure, stroke sequencing)	✗	✓ Limited	✗	✓	✗	✓	✗	✗	✓
Game-based assessment tasks	✓	✓ Limited	✗	✓	✓	✗	✗	✓	✓
Aimed at therapeutic intervention (treatment)	✓ Limited	✗	✗	✗	✗	✗	✓	✓	✓
Continuous progress monitoring & dashboards	✗	✗	✗	✗	✗	✗	✗	✓ Limited	✓

Table 01 - System Comparison

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate an integrated, game-based, rule-driven mobile handwriting intervention system specifically for Sinhala-speaking children aged 6-8 years with dysgraphia. The system will objectively assess handwriting difficulties including poor letter formation, inconsistent spacing and alignment errors through initial severity classification. Deliver adaptive intervention activities including both digital games and sensory-based learning tasks with voice-guided instructions in Sinhala. And provide continuous progress monitoring with stakeholder dashboards for parents, teachers, and therapists, all to be developed and evaluated within a defined development period.

2.2 Specific Objectives

1. **Initial Level Identification** - Children begin with handwriting games designed for lower levels and gradually progress through the activities. The system identifies the stage at which they experience difficulty completing tasks.
2. **Develop Adaptive Game-Based Intervention Activities** - Design engaging, child-friendly digital handwriting games that both evaluate and strengthen handwriting skills. Activities will automatically adjust difficulty parameters such as letter complexity, tracing guidance, spacing support, and time limits based on the child's performance trends, error patterns, and severity level, ensuring personalized and adaptive remediation.
3. **Sensory-Based Learning Integration** - The system integrates sensory-based learning with a hybrid handwriting verification approach by implementing multisensory writing activities, such as writing letters in sand or on textured surfaces, to strengthen motor-memory connections and enhance letter recognition and handwriting fluency. These activities are supported by an image-based verification mechanism that combines OpenCV for image preprocessing and a ML model for letter- and word-level classification, while leveraging MyScript for accurate word-level correctness validation.
4. **Voice-Guided Instructions** - Provide clear, child-friendly audio guidance in Sinhala to help early learners (6–8 years) understand task instructions without relying solely on reading ability, enhancing accessibility and engagement.
5. **Personalized Game Recommendation** - Develop a rule-based scoring algorithm that analyzes handwriting performance metrics including accuracy, completion time, spacing errors, and alignment deviations and, based on predefined threshold criteria, recommends appropriate intervention games tailored to the child's identified difficulty level.
6. **Develop continuous performance tracking and reporting dashboard** - Build a progress monitoring system that captures longitudinal handwriting data, visualizes improvement trends across multiple sessions, and generates simplified, non-technical progress reports for parents and teachers to track remediation outcomes and adjust learning strategies.

3. METHODOLOGY

Research approach

This research adopts a Design Science Research (DSR) methodology combined with an experimental evaluation approach to design and validate a game-based mobile handwriting intervention system for children with dysgraphia.

The study follows stages including problem identification, requirement analysis, system design, development, and evaluation. Initially, the research analyzes the handwriting difficulties faced by children with dysgraphia and identifies limitations of traditional handwriting assessment methods. Based on these findings, a game-based adaptive mobile system is designed to assess handwriting skills and provide personalized intervention for children aged 6-8 years.

The system uses game-based handwriting assessment tasks and a rule-based algorithm to analyze stroke accuracy, letter formation, spacing, alignment, letter mirroring, and writing speed. It classifies the child's difficulty level and provides adaptive learning activities tailored to their needs. Sensory-based tasks like writing in sand strengthen motor-memory connections, while voice-guided instructions in Sinhala help young children understand tasks. Progress dashboards enable parents, teachers, and therapists to track improvement and intervention effectiveness, offering a comprehensive, personalized platform for Sinhala-speaking children with dysgraphia.

The system is developed using an Agile iterative software development approach, allowing continuous improvement through repeated cycles of design, implementation, testing, and refinement.

Technical methods and algorithms

Digital ink capture forms the foundation of data collection, where every stroke drawn on the touchscreen is recorded as a sequence of spatial coordinates with corresponding timestamps. This raw data preserves the complete writing trajectory, enabling detailed analysis of how letters are formed rather than merely what is written.

Letter size analysis involves calculating bounding box dimensions for each character and establishing statistical distributions of height and width across writing samples to identify inconsistencies or abnormal proportions.

Spacing analysis examines both within-word character distances and between-word gaps, quantifying consistency through statistical measures. Baseline adherence is evaluated by fitting regression lines through character bottom coordinates and measuring vertical deviations.

The system combines outputs from the machine learning model and MyScript recognition to improve verification accuracy. The machine learning model performs letter-level classification and evaluates handwriting quality, while MyScript validates word-level correctness. In cases where both outputs agree, the result is considered highly reliable. If discrepancies occur,

additional rule-based scoring based on stroke patterns and spacing is applied as a fallback mechanism.

Voice-guided instruction generation involves converting textual prompts into audible speech using a Text-to-Speech (TTS) engine. The system delivers instructions dynamically to guide the child through handwriting tasks, ensuring clear, consistent, and engaging directions during each activity.

For sensory-based handwriting tasks, such as writing letters in sand, the system captures the child's writing using a tablet or phone camera under good lighting and contrast. Images are preprocessed with OpenCV-converted to grayscale, thresholded, and edge-detected-to extract letter contours. The processed output is then passed to the machine learning model for classification.

A weighted scoring algorithm integrates multiple data sources to classify dysgraphia severity into mild, moderate, or severe levels. The scoring model combines parent questionnaire responses, which capture observational insights about the child's writing behavior with directly measured handwriting performance metrics from the assessment tasks.

The adaptive game recommendation engine operates through a rule-based decision matrix that considers the child's identified severity level, specific error patterns observed during assessment, and ongoing performance trends during intervention activities.

Tools and platforms with justification

The frontend application is developed using React, selected for its component-based architecture that supports reusable UI elements across multiple game modules and efficient rendering of dynamic handwriting canvases. React enables the development of interactive, animated game activities, enhancing child engagement through smooth transitions and real-time visual feedback.

Handwriting capture is implemented using HTML5 Canvas in conjunction with React-Konva, which provides a lightweight and performant drawing surface capable of recording detailed stroke data. This ensures precise, real-time handwriting data collection while maintaining smooth rendering for user interaction.

The voice-guided instructions are implemented using a TTS engine to provide clear, dynamic, and child-friendly guidance in Sinhala, addressing limited reading abilities of young learners. The sensory-based handwriting verification employs template matching with OpenCV to analyze letter contours captured from sand writing or similar tactile activities. This approach is lightweight, computationally efficient, and provides objective, quantitative feedback on letter shape, spacing, and alignment.

Machine learning model development is carried out using frameworks such as TensorFlow or PyTorch. These tools are used to train and evaluate handwriting classification models based on labeled Sinhala handwriting datasets.

The backend architecture utilizes Node.js with Express, chosen for its event-driven, non-blocking nature that efficiently handles concurrent game sessions and real-time data processing. The consistent JavaScript environment across frontend and backend streamlines development and maintenance.

A dual-database strategy is adopted:

- **MongoDB** stores user profiles, game configurations, and aggregated performance metrics, benefiting from its flexible document-based schema.
- **Firebase** supports real-time synchronization of active game sessions and progress tracking. Firebase Authentication is utilized for secure user management.

This architecture combines flexible schema design through MongoDB with real-time capabilities provided by Firebase, ensuring scalability and cost-effectiveness.

Version control is managed using Git, with the project repository hosted on GitHub. The development environment uses Visual Studio Code (VS Code) as the integrated development environment (IDE) for coding, debugging, and project management.

Data collection and ethical considerations

The data collection process focuses on gathering expert knowledge from healthcare professionals and collecting handwriting performance data related to dysgraphia characteristics, rather than obtaining sensitive personal information from children. This approach minimizes ethical complexity while ensuring clinical validity. Data collected includes handwriting indicators such as letter formation, spacing consistency, alignment precision, letter mirroring, and writing fluency, alongside expert input on severity levels categorized as mild, moderate, or severe. To design assessment tasks, parent questionnaires, and adaptive game activities, private pediatricians and educational therapists were consulted to identify key handwriting difficulties, appropriate assessment questions, and effective intervention exercises.

In addition, handwritten samples of Sinhala letters and words will be collected from primary school students to support the training of the machine learning model used for handwriting classification. These samples will be anonymized and collected with proper consent from schools and guardians, ensuring that no personally identifiable information is stored. The collected dataset will be used to train and validate the model to accurately recognize variations in children's handwriting and improve the reliability of the verification process.

Validation strategy and performance metrics

After developing the system, validation will be conducted with a small group of 5-6 children to evaluate technical performance, usability, and intervention effectiveness. Technical validation will check the accuracy of handwriting capture, feature extraction, scoring, and adaptive game recommendations against expert observations. Pre- and post-intervention comparisons will assess improvements in handwriting metrics such as letter formation, spacing, alignment, and fluency over the target intervention period to determine whether children will show measurable progress.

The performance of the machine learning model will be evaluated using metrics such as accuracy, precision and recall. Additionally, the effectiveness of the hybrid verification approach will be assessed by comparing results from the machine learning model, MyScript recognition, and combined outputs against expert evaluations.

Usability will be evaluated through child engagement observations, parent feedback, and System Usability Scale scores. Performance targets will include classification accuracy above 80%, strong agreement with expert assessments, and observable improvement in handwriting skills alongside high child engagement throughout the intervention.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed solution is an integrated, game-based, adaptive handwriting intervention system for Sinhala-speaking children aged 6-8 with dysgraphia. It assesses handwriting, tracks progress via therapist-friendly dashboards, and combines sensory-based activities, digital ink capture, handwriting verification, adaptive game recommendations, and real-time reporting. Voice-guided instructions in Sinhala support engagement and accessibility for children with limited reading skills.

The system begins with the child performing assessment and sensory-based handwriting activities, during which digital ink strokes and images are captured. These inputs are analyzed using multiple components: a feature extractor evaluates metrics such as accuracy, spacing, alignment, mirroring, and fluency, MyScript verifies word correctness, and OpenCV is used for preprocessing sensory-based handwriting inputs, while the machine learning model performs the final classification and verification. Based on this analysis, the system classifies the child's handwriting difficulty as mild, moderate, or severe. A rule-based engine then selects adaptive games tailored to the child's level. All performance metrics are securely stored in the backend, and parents or teachers can review progress through dashboards.

The system is designed for scalability and security. Event-driven Node.js and a modular frontend/backend architecture support multiple users and allow easy addition of new games or assessment modules. Security is ensured through encrypted communication, secure authentication, API access control, and minimal storage of sensitive data, while cloud services such as MyScript and Firebase use secure protocols to protect user information.

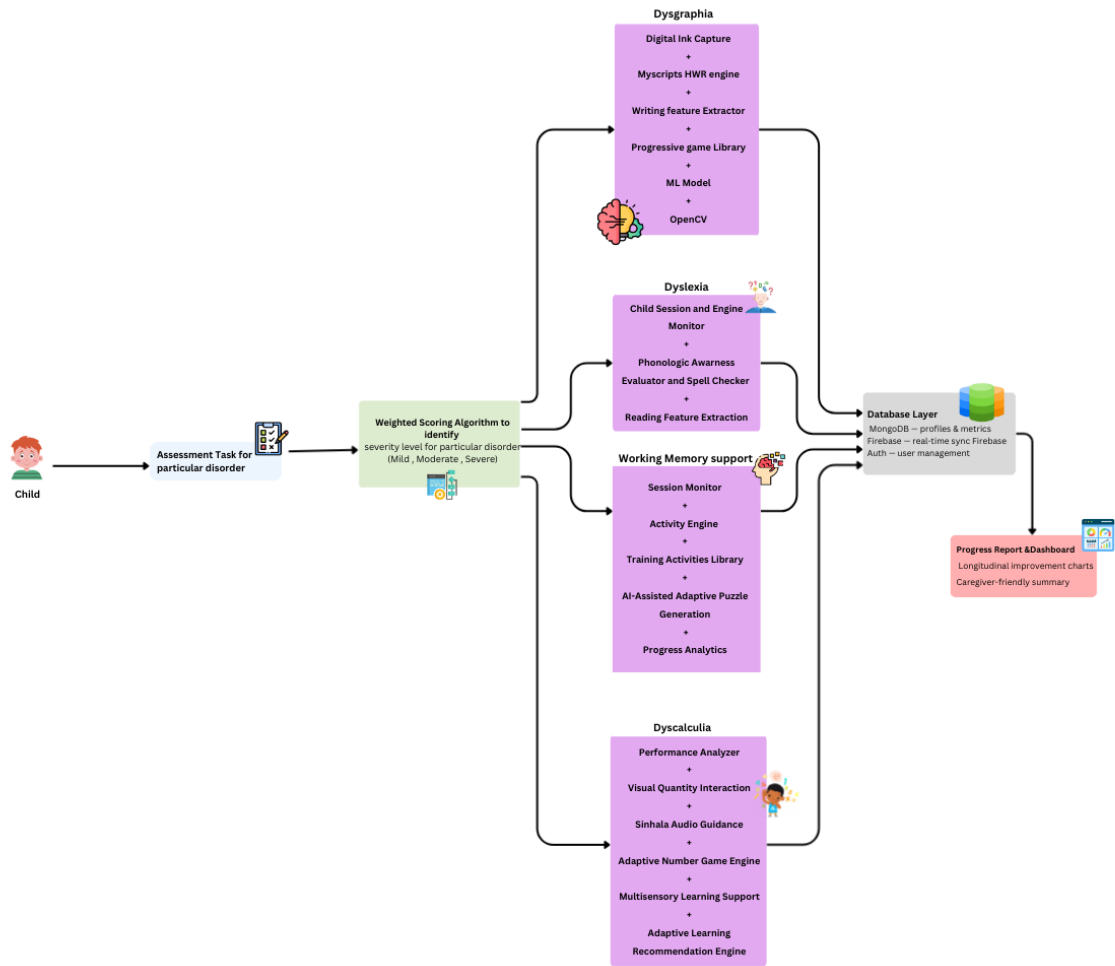


Figure 01 - System Diagram

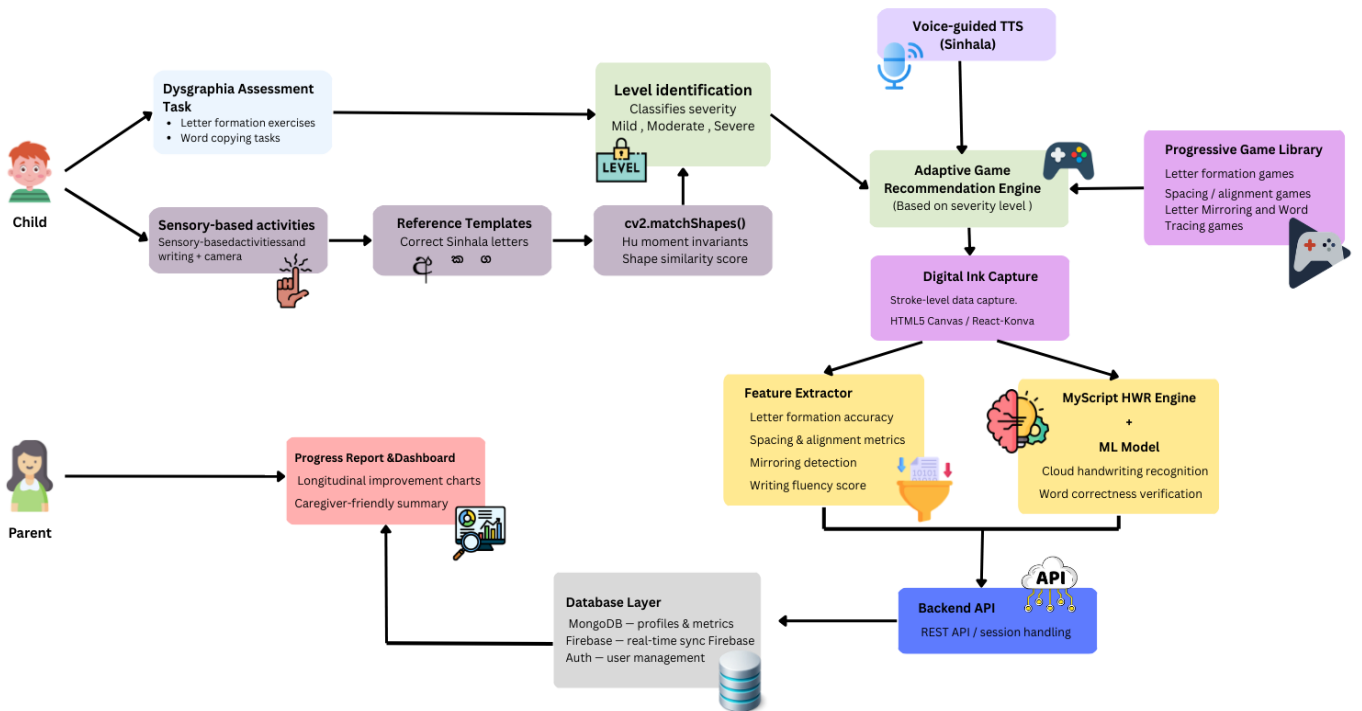


Figure 02 - Component diagram for dysgraphia intervention module

5.USER REQUIREMENTS

Stakeholder identification

Primary Users: Children aged 6-8 years with handwriting difficulties (dysgraphia).

Secondary Users: Therapist, Parents and teachers who will Responsible for creating child profiles, monitoring progress reports, and supervising system usage.

Tertiary Users: Clinical experts who provide input on assessment design, validate system classifications, and may use system-generated reports to support clinical decision-making. Educational professionals may also refer children for screening and use progress reports to inform classroom support strategies.

Functional requirements

01. **User Registration and Login** - The system shall allow guardian to register and securely log in using email and password authentication to access the platform.
02. **Handwriting Assessment Tasks** - Children complete interactive, gamified handwriting tasks that evaluate letter formation, spacing, alignment, mirroring, and writing fluency.
03. **Dysgraphia Level Classification** - The system shall begin with handwriting games designed for lower difficulty levels and gradually progress through more advanced activities. It shall identify the stage at which the child experiences difficulty in completing tasks to classify their dysgraphia severity level.
04. **Sensory-Based Handwriting Verification** – The system shall capture handwriting images, preprocess them using OpenCV techniques, and classify them using a machine learning model to determine correctness.
05. **Voice-Guided Instructions** - The system shall provide audible, child-friendly instructions in Sinhala using a TTS engine to guide children through handwriting tasks.
06. **Adaptive Game Recommendation** - The system shall recommend appropriate intervention games based on the child's identified severity level.
07. **Digital Ink Capture** - The system shall record every stroke drawn on the touchscreen as sequences of spatial coordinates with corresponding timestamps to preserve complete writing trajectory data.
08. **Word Correctness Verification** - The system shall integrate MyScript handwriting recognition technology to compare child's written words against expected words for activities requiring spelling accuracy assessment.

09. **Progressive Game Library** - The system shall maintain a library of handwriting intervention games with adjustable difficulty parameters including letter complexity, tracing guidance level, spacing support, and time constraints.
10. **Real-Time Performance Tracking** - The system shall log child performance during game activities including accuracy, completion time, error types, and engagement metrics for continuous progress monitoring.
11. **Parent-Friendly Report Generation** - The system shall generate comprehensive progress reports for guardians summarizing the child's handwriting performance, identified difficulties, severity level, and improvement trends over time.

Non-functional requirements

1. **Performance** - The system shall capture and render handwriting strokes in real time with minimal latency to ensure a natural writing experience. Feature extraction and performance metrics shall be generated within four seconds of task completion. Game animations and transitions shall maintain smooth performance at a minimum of 30 frames per second to support child engagement.
2. **Reliability** - Data should be logged consistently across sessions, with offline caching and automatic upload when connectivity is restored.
3. **Scalability** - The database architecture shall support scalability to store data for at least 1,000 child profiles without performance degradation. The system shall handle increasing numbers of concurrent game sessions through cloud-based infrastructure capable of scaling on demand.
4. **Usability** - The child interface shall use bright colors, friendly animations, and age-appropriate visuals to maintain engagement for children aged 6-8 years. The system shall provide visual and audio feedback upon task completion to reinforce positive engagement. The parent interface shall present assessment results in clear, non-technical language with supportive visual aids to ensure easy understanding.
5. **Security and Privacy** - Child data must be anonymized and encrypted. only authorized caregivers and clinicians can access reports.
6. **Maintainability** - The system architecture shall be modular, with clear separation between frontend, backend, and database layers to support independent updates and maintenance. The codebase shall be well-documented and adhere to standard development practices to ensure that future developers can easily understand, maintain, and extend the system. New game activities can be added to the game library without modifying the core architecture, ensuring scalability and extensibility.

6. COMMERCIALIZATION PLAN

Target market and customer persona

The primary target market for the proposed system includes parents, therapists, and primary schools supporting children aged 6–8 years who experience handwriting difficulties or dysgraphia. The application is particularly valuable for Sinhala-medium educational environments, where language-specific intervention tools are limited.

1. Child User Persona

Name	Isuri Piyumali
Age	7 years
Background	Isuri enjoys playing games on tablets but finds writing activities difficult and frustrating. He often struggles with forming letters correctly and maintaining proper spacing between words.
Goals	Learn to write letters correctly. Complete writing tasks without frustration. Enjoy learning activities.
Challenges	Difficulty remembering correct letter shapes. Low confidence when writing. Gets bored with repetitive writing exercises.
How the System Helps	The application uses game-based learning, voice instructions, and interactive, sensory-based handwriting activities, making writing practice more engaging and enjoyable for children like Isuri.

2. Therapist Persona

Name	Dr. Sanduni Jayasinghe
Age	40 years
Occupation	Occupational Therapist
Background	Dr. Sanduni works with children who have learning difficulties such as dysgraphia. She conducts handwriting therapy sessions but often struggles to track children's practice outside the clinic.
Goals	Provide structured handwriting exercises for children Monitor improvement in handwriting performance Recommend appropriate intervention activities

Challenges	Limited time to monitor each child's progress between therapy sessions. Lack of digital tools for Sinhala handwriting analysis. Difficulty engaging children with traditional writing exercises.
How the System Helps	The system provides data-driven handwriting analysis, progress dashboards, and adaptive learning activities, allowing therapists to monitor improvements and recommend personalized interventions.

3. Parent Persona

Name	Nadeesha Perera
Age	35 years
Background	Nadeesha is a working mother with a 7-year-old child who struggles with handwriting tasks at school. She notices that her child avoids writing activities and becomes frustrated during homework. She is looking for a simple digital tool that can help improve her child's handwriting skills at home.
Goals	Help her child improve handwriting and writing confidence. Monitor the child's progress over time. Find an affordable alternative to frequent therapy sessions.
Challenges	Limited knowledge about dysgraphia and handwriting difficulties. Lack of accessible learning tools in Sinhala. Limited time to supervise long practice sessions.
How the System Helps	The application provides interactive handwriting games, progress reports, and voice-guided activities that help the child practice independently while allowing the parent to track improvement.

Value proposition

Value Element	Description
Affordable Intervention	Reduces need for expensive ongoing therapy sessions by providing daily practice tool at fraction of cost.
Clear Progress Visibility	Caregiver-friendly reports showing child's improvement in letter formation, spacing, and writing fluency.
Inclusive Classroom Support	Enables teachers to support students with handwriting difficulties without specialized training.
Time Savings	Reduces time spent on manual progress documentation and parent updates.
Engaging Game Experience	Writing practice disguised as fun, interactive games with animations and rewards.
Builds Confidence	Positive reinforcement and visible progress reduce writing-related frustration.
Age-Appropriate Design	Bright visuals, friendly characters, and audio prompts designed for 6-8-year-olds.
Peace of Mind	Confidence that child is receiving targeted support for identified difficulties.

Table 02 - value proposition

Revenue model

The proposed handwriting intervention system can adopt a freemium-based, multi-tier revenue strategy, ensure accessibility while generate sustainable revenue. The system targets parents, therapists, and educational institutions, providing engaging, language-specific handwriting support enhanced with voice-guided instructions and sensory-based writing activities.

Freemium Model: Basic handwriting games and limited progress tracking available for free to attract users.

Subscription Model: Paid plans unlock adaptive games, detailed analytics, personalized dashboards, voice-guided instructions, and sensory-based tasks.

Institutional Licensing: Schools and therapy centers get multi-user access, dashboards, and full system integration.

B2B Partnerships: Collaborate with pediatric clinics and educational organizations for therapy and learning programs.

Cost estimation

Cost Category	Estimated Cost (LKR)	Details
Software Development	Rs. 35,000 - 50,000	React frontend, Node.js backend, database integration.
Game Design	Rs. 10,000 - 15,000	Child-friendly animations, characters, UI elements for game activities
Expert Consultation	Rs. 10,000 - 15,000	Compensation for occupational therapists and pediatric specialists for validation
MyScript Integration	Rs. 0 - 10,000	Free tier available, minimal during development
Testing and Piloting	Rs. 10,000 - 15,000	Recruitment of participants, incentives, usability testing.
TOTAL INITIAL COSTS	Rs. 65,000 - 100,000	

Table 03 - Cost Estimation

Pricing strategy

The system adopts an affordable and accessible pricing approach to ensure widespread adoption among families, schools, and therapy centers while maintaining sustainability.

Penetration Pricing: Initial low-cost subscriptions to attract early users and build a user base.

Individual Plans:

- **Basic Plan:** Free or low-cost monthly access to core handwriting games and limited progress tracking.
- **Standard Plan:** Full access to adaptive intervention games, voice-guided instructions, sensory-based activities, and comprehensive progress monitoring dashboards.

Institutional Plans: Discounted packages for schools, therapy centers, and educational institutions, enabling multi-student access and administrative dashboards.

Promotional Trials: Free trials to encourage initial adoption and allow users to experience adaptive and sensory-based features before subscribing.

This strategy balances affordability, value delivery, and scalability, ensuring accessibility in low-resource settings while providing a sustainable revenue model for long-term operation.

Competitive advantage

The proposed system offers several competitive advantages. It is specifically designed for Sinhala-language learners, addressing a major gap in existing handwriting intervention tools. The integration of game-based learning, adaptive difficulty adjustment, sensory-based writing verification, and voice-guided instructions enhances user engagement and learning effectiveness. Additionally, the system provides real-time progress tracking and personalized recommendations, which are not commonly available in standard handwriting practice applications in Sri Lanka.

Intellectual property considerations

The project may involve protecting the software design, algorithms, and educational content used in the platform. The system can be protected through copyright registration for the software code and educational game content. If the handwriting analysis algorithms or adaptive recommendation methods demonstrate novelty, they may also be considered for patent protection. Furthermore, the application name and logo can be protected through trademark registration to support future commercialization.

7.BUDGET AND JUSTIFICATION

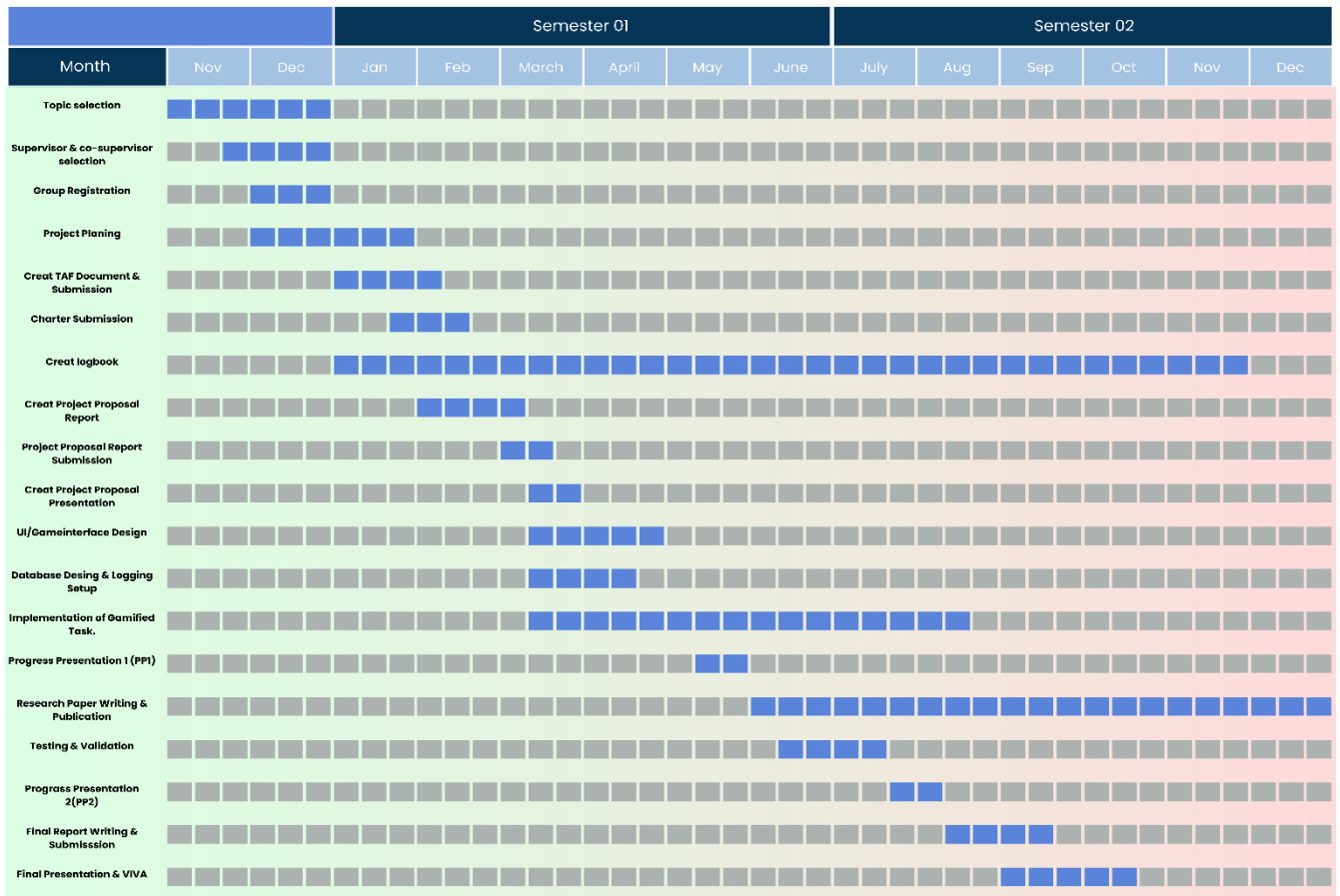
Cost Category	Estimated Cost (LKR)	JUSTIFICATION
Travel expenses	9,000	Visits to the SLIIT Matara Center to meet the Supervisor and Co-supervisor. Visit to Ayati Center and Other Disability Treatment Centers.
Data collection	15,000	Arrange private appointments to collect data and expert insights regarding dysgraphia.
Printing & documentations	1,500	Printing proposals, reports, consent sheets and questionnaires.
Total	25,500	

Table 04 - Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase	Task	Timeline
Project Planning & Requirement Analysis	Literature review and problem identification	Week 1 – Week 4
	Define system requirements and functional specifications	Week 5 – Week 6
	Design research methodology and technical approach	Week 7 – Week 8
	Prepare project proposal documentation	Week 9- week10
System Design	System architecture and workflow design	Week 10 – Week 11
	Database schema design (MongoDB, Firebase)	Week 12
	UI/UX design for handwriting games and dashboards	Week 13 – Week 14
Frontend Development	Develop login, guardian profile, and questionnaire modules	Week 15 – Week 17
	Implement handwriting canvas using React-Konva	Week 19 – Week 20
	Develop interactive handwriting games and UI animations	Week 21 – Week 23
	Integrate voice-guided instructions using TTS	Week 24
Backend Development	Implement API endpoints and database integration	Week 26 – Week 27
	Implement rule-based scoring algorithm and dysgraphia classification	Week 28
Advanced Features	Integrate MyScript handwriting recognition API	Week 29
	Implement sensory-based handwriting verification using OpenCV	week 30 – week 31
	Develop progress monitoring dashboards	Week 32
System Testing & Validation	Functional testing of modules	Week 33
	Conduct usability testing and analyze performance results	Week 33 – Week 34
Documentation & Finalization	Prepare final research report, system documentation, and presentation materials	Week 35 – week 36

9. GANTT CHART



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11. APPENDICES

Survey instruments

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders

Introduction

Smart Learn: An Adaptive Mobile Platform for Neurodevelopmental Learning Disorders is a research and development project conducted by final-year undergraduate students in Information Technology. The project focuses on designing a supportive digital learning application for children with neurodevelopmental learning difficulties such as dyslexia, dyscalculia, dysgraphia, and working memory challenges.

The purpose of this questionnaire is to gather professional insights from therapists, educators, and specialists about current therapy activities, assessment methods, and effective learning strategies for children aged approximately 6-8 years.

The information collected will be used only for academic research and application development purposes. It will help us design a more effective, adaptive, and child-friendly learning application. All responses will be kept confidential and will only be used to improve the quality, usability, and effectiveness of the proposed mobile platform.

Questions

1. What types of activities or exercises are currently used in therapy for children with dyslexia, dyscalculia, dysgraphia, and working memory difficulties?
2. Are these activities usually paper-based, game-based, digital, or therapist-guided?
3. Which activities do you find most effective for children aged 6-8?
4. How do these activities change as the child improves or struggles?
5. Is there any recognized way to categorize these disorders into levels such as mild, moderate, or severe in clinical practice?
6. Do therapists assess children based on functional ability levels rather than biological markers?
7. How do you typically identify a child's current level - through standardized tests, observation, questionnaires, or activities?
8. If an app were to adjust its difficulty level based on a child's performance, what factors should it consider to avoid frustration or overstimulation?
9. What signs indicate a child should move to a higher or lower level?
10. Are there any activities or approaches that should be avoided for children with these disorders?
11. What are the signs that a child is overwhelmed rather than challenged?
12. What kind of progress information is most useful for parents?
13. How do therapists usually communicate improvement to parents?
14. From your professional experience, what is one feature you wish learning apps for neurodevelopmental disorders did better?

Risk analysis

A specific risk matrix evaluating the potential failures of the mobile handwriting intervention system, including the handwriting capture module, adaptive scoring engine, and backend services.

Risk Factor	Impact	Probability	Mitigation Strategy
Inaccurate Handwriting Recognition.	High	Medium	Use rule-based validation together with MyScript results. If the recognition confidence is low, the system will analyze stroke patterns instead of relying only on word recognition.
Incorrect Severity Classification.	High	Low	Use expert-defined threshold values and test the scoring algorithm with sample handwriting data. The system will also log classification results for expert review and improvement.
Touchscreen Input Noise (Children draw unintended strokes or accidental taps)	Medium	High	Remove very short strokes and accidental taps, and apply a smoothing algorithm to clean the handwriting data before analysis.
Data Privacy and Security Risks.	Critical	Low	Use secure login, HTTPS encryption, and anonymized data storage to protect children's handwriting data. Only authorized caregivers can access the reports.
MyScript service temporarily unavailable	Medium	Low	Implement fallback evaluation using internal rule-based handwriting metrics so the system can continue functioning even if the external API is unavailable.

Plagiarism Report

Proposal IT22128836.docx

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